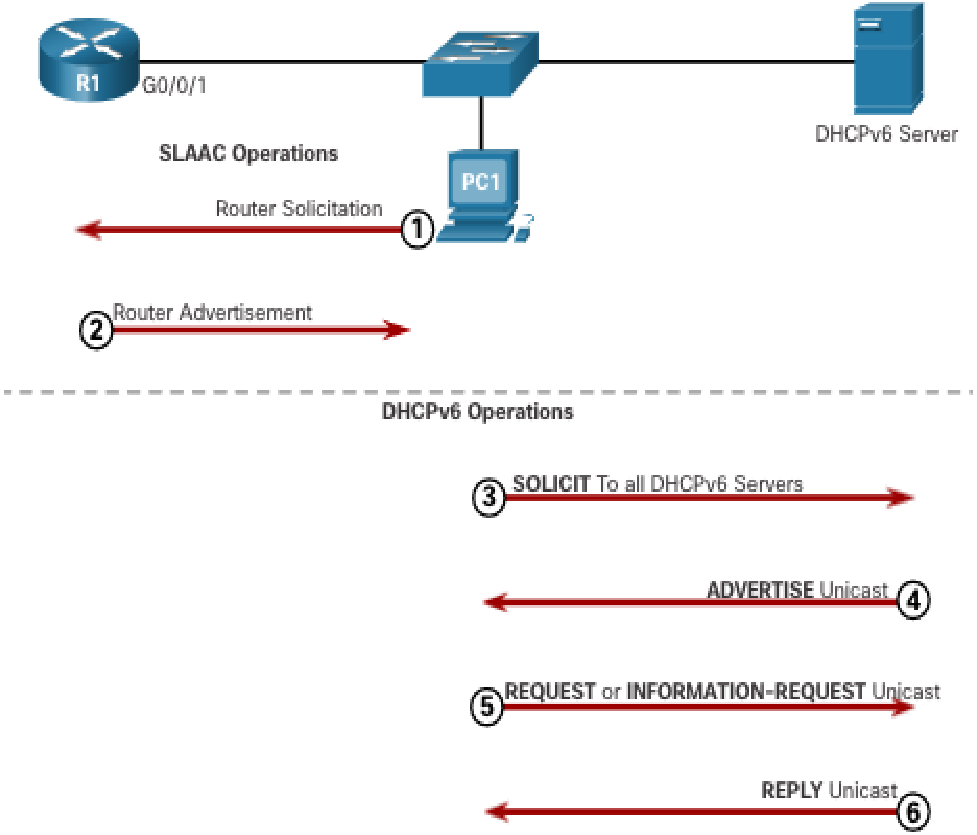
8.3 DHCPv6

DHCPv6 Operation Steps

Regardless, when an RA indicates to use DHCPv6 or stateful DHCPv6:

1. The host sends an RS message.
2. The router responds with an RA message.
3. The host sends a DHCPv6 SOLICIT message.
4. The DHCPv6 server responds with an ADVERTISE message.
5. The host responds to the DHCPv6 server.
6. The DHCPv6 server sends a REPLY message.

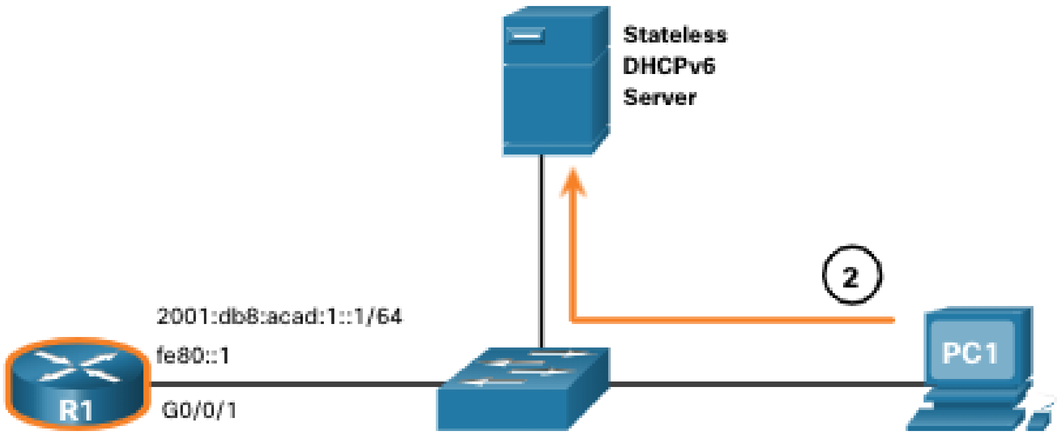


DHCPv6

Stateless DHCPv6 Operation

For example, PC1 receives a stateless RA message containing:

- The IPv6 GUA network prefix and prefix length.
- A flag set to 1 informing the host to use SLAAC.
- O flag set to 1 informing the host to seek that additional configuration information from a DHCPv6 server.
- M flag set to the default value 0.



Flag	Value
A	1
O	1
M	0

PC1 sends a DHCPv6 SOLICIT message seeking additional information from a stateless DHCPv6 server

Enable Stateless DHCPv6 on an Interface

Stateless DHCPv6 is enabled using the

ipv6 nd other-config-flag

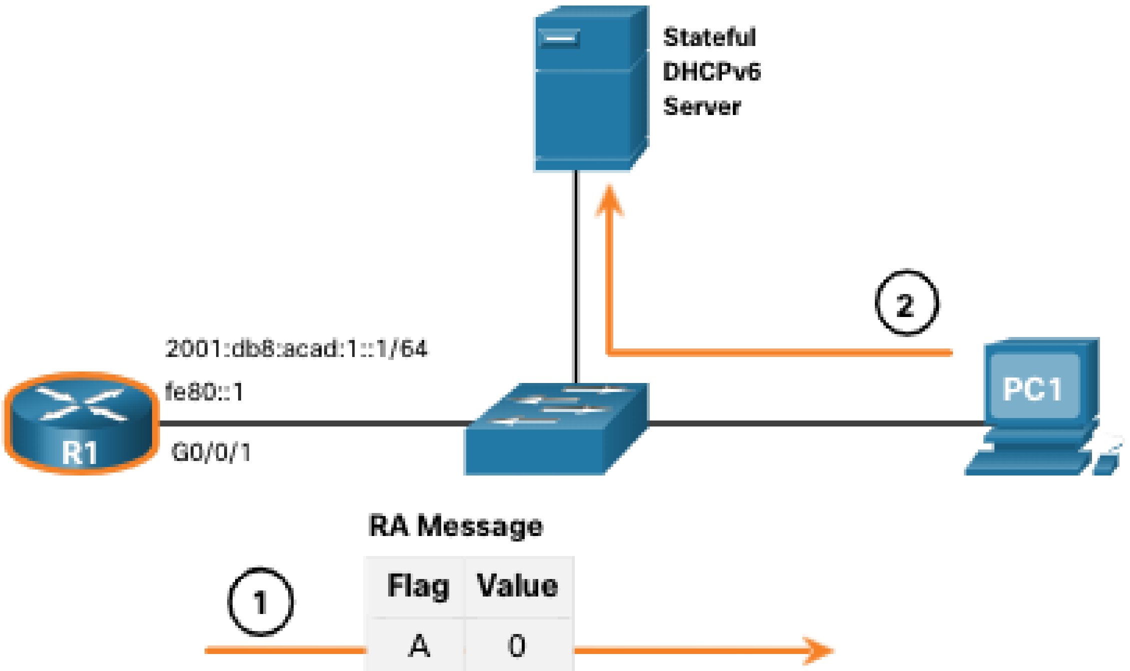
interface configuration command setting the O flag to 1.

```
R1(config-if)# ipv6 nd other-config-flag
R1(config-if)# end
R1#
R1# show ipv6 interface g0/0/1 | begin ND
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND advertised reachable time is 0 (unspecified)
  ND advertised retransmit interval is 0 (unspecified)
  ND router advertisements are sent every 200 seconds
  ND router advertisements live for 1800 seconds
  ND advertised default router preference is Medium
  Hosts use stateless autoconfig for addresses.
  Hosts use DHCP to obtain other configuration.
R1#
```

Stateful DHCPv6 Operation

If an RA indicates the stateful DHCPv6 method, the host contacts a DHCPv6 server for all configuration information.

- **Note**: The DHCPv6 server is stateful and maintains a list of IPv6 address bindings.



RA Message

Flag	Value
A	0
O	0
M	1

PC1 sends a DHCPv6 SOLICIT message seeking additional information from a stateful DHCPv6 server.

Enable Stateful DHCPv6 on an Interface

Stateful DHCPv6 is enabled using the

ipv6 nd managed-config-flag

interface configuration command setting the M flag to 1.

The highlighted output in the example confirms that the RA will tell the host to obtain all IPv6 configuration information from a DHCPv6 server (M flag = 1).

```
R1(config)# int g0/0/1
R1(config-if)# ipv6 nd managed-config-flag
R1(config-if)# end
R1#
R1# show ipv6 interface g0/0/1 | begin ND
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND advertised reachable time is 0 (unspecified)
  ND advertised retransmit interval is 0 (unspecified)
  ND router advertisements are sent every 200 seconds
  ND router advertisements live for 1800 seconds
  ND advertised default router preference is Medium
  Hosts use DHCP to obtain routable addresses.
R1#
```